

Mohammad Ghaffary

Senior Backend Engineer · Python · Distributed Systems

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SUMMARY

Senior Backend Engineer with 8 years of experience designing, scaling and modernizing Python systems. Most recently at **Nobitex, Iran's largest cryptocurrency exchange (10M+ users)**, where I re-architected coin and blockchain-network integration and led the safe migration of money-handling code off a legacy stack. Hands-on with **Django/DRF, FastAPI, event-driven microservices with Kafka and RabbitMQ, PostgreSQL performance tuning, Docker and Kubernetes**. I like turning slow, risky manual processes into automated, well-tested ones, and I mentor other engineers along the way.

SKILLS

Languages	Python (expert), SQL, Bash; familiar with TypeScript
Backend	Django, Django REST Framework, FastAPI, Celery, SQLAlchemy 2.x, Pydantic v2, REST API design, OpenAPI/Swagger
Distributed systems	Microservices, Apache Kafka, RabbitMQ, event-driven architecture, idempotent producers/consumers, concurrency control
Data	PostgreSQL (query tuning, indexing, row-level locking), Redis, MongoDB, ClickHouse, Alembic, ETL pipelines
Cloud & DevOps	Docker, Docker Compose, Kubernetes, CI/CD, Linux, Prometheus, Grafana
Practices	Testing with pytest (TDD), SOLID, design patterns, legacy modernization, code review, mentoring, Agile/Scrum
Automation & AI	n8n (custom community nodes), LLM API integration, Telegram bots

EXPERIENCE

Freelance Backend Engineer — QueueLess (Remote) 2026 – Present

Event-driven SaaS that replaces physical lines at clinics, barbers, restaurants and public offices with virtual tickets, live position tracking and notifications.

- Designed the target architecture as independent microservices (Identity, Queue, Business, Notification, Waiting-Time, Analytics) with a database per service, Kafka for domain events and synchronous HTTP for queries.
- Built the core Queue Service with FastAPI, SQLAlchemy 2.x, Pydantic v2 and PostgreSQL. Ticket issuing and “call next” run under row-level locks (`SELECT ... FOR UPDATE`), so concurrent requests never produce duplicate or skipped tickets.
- Modeled the ticket lifecycle as a state machine enforced by atomic compare-and-set SQL updates and partial unique indexes: when two actions race on one ticket, exactly one wins and the other receives 409.
- Published domain events to Kafka (KRaft) with idempotent producers, `queue_id` partition keys for per-queue ordering, and a versioned event envelope that lets consumers deduplicate and evolve schemas.
- Containerized the stack (health-gated startup, one-shot migration and topic-init jobs), ready for Kubernetes deployment with health probes and environment-based config, and covered race conditions and event publishing with pytest.

Tech: Python, FastAPI, Kafka, PostgreSQL, SQLAlchemy, Alembic, Docker, Kubernetes, pytest · github.com/sheracore/queueless

Senior Software Engineer — Nobitex (Tehran, Iran) Aug 2023 – May 2026

Iran's largest cryptocurrency exchange: 10M+ users, high-volume trading, deposits and withdrawals across dozens of blockchains.

- Re-architected cryptocurrency coin and network integration to be configuration-driven, cutting the time to add a new coin or network from **~3 hours to ~5 minutes** with **no code deployment**, and untangled the system-wide dependencies that had made it hard-coded.

- Built a compatibility layer that translated responses from the new Django REST Framework APIs into legacy formats, letting the team migrate endpoints incrementally with **zero breaking changes for clients**, plus centralized logging and error handling.
- Designed a deposit-simulation harness that ran the same scenarios through legacy and refactored code and logged any diverging responses. This shadow testing of money-handling code made the refactoring **~20% faster** and caught regressions before release.
- Mentored a mid-level engineer through the delivery of a Telegram bot that lets product owners list new blockchain tokens themselves, reducing engineering effort for token listings by **~90%**.
- Profiled and tuned slow PostgreSQL queries and ORM access patterns on critical paths (indexes, query rewrites, eliminating N+1 queries), reducing database hits by **up to 50%**.

Tech: Python, Django, DRF, Celery, RabbitMQ, PostgreSQL, Redis, Docker, Prometheus, Grafana

Backend Engineer (Django) — Learnwise (Remote)

Jul 2022 – Aug 2023

Misalearn, a language-learning application.

- Cut client requests for Oxford dictionary updates by **95%** by replacing per-word polling with periodically generated, encrypted JSON bundles built by Celery Beat.
- Built a reusable Django module for metadata generation, validation and key control, adopted across the application by other modules.
- Reached **100% API documentation coverage** by extending Swagger/OpenAPI schema generation to support custom field types and django-filter endpoints.
- Developed and published an n8n community node, and integrated LLM-generated content into the product's learning flows.

Tech: Python, Django, DRF, Celery, PostgreSQL, Redis, Swagger/OpenAPI, n8n

Backend Engineer & Data Analyst — Farafan (Tehran, Iran)

Jun 2018 – Jun 2022

Telecommunications and IT solutions provider (300+ employees).

- **VirusPaad (SaaS):** built the Django backend of a multi-engine file-scanning platform, available on the web and on kiosks, that checks files against 40+ antivirus engines. Orchestrated 40+ VMs through RPyC remote execution and ingested files over Samba and FTPS.
- **Radio Patrol (telecom analytics):** automated **~80%** of manual reporting with ETL pipelines that extract large-scale mobile network data from ClickHouse, compute KPIs into PostgreSQL and visualize them in Grafana.

Tech: Python, Django, RPyC, ClickHouse, PostgreSQL, Grafana, Linux

PROJECTS

JobFinder — Visa-sponsorship job aggregator for Europe. Collects postings from job boards and company career APIs (Greenhouse, Lever, Ashby), detects sponsorship and relocation signals, cross-checks official sponsor registers and ranks roles against a skills profile. Python, httpx, SQLite, GitHub Actions · github.com/sheracore/JobFinder

EDUCATION

M.Sc. in Information Technology Engineering — K. N. Toosi University of Technology, Tehran

LANGUAGES

English: professional working proficiency · Persian: native